The NULLx Learning Kit

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Presented by:

Members: Meshu Deb Nath
Diwakar Dangal
Rakibul Islam

AGENDA

- ➢ Who We Are
- ➢ Purpose
- ➢ Problem Definitions
- ➢ Survey and their Results
- ➢ Solution
- ➢ Product
- ➢ Demonstration

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Who we Are?

We **NULLx** are an online Teaching and learning platform that is dedicated to delivering the best teaching solution to our customers based on Microservice patterns.

- We are providing education on Microservices pattern and supporting procedures through a WebShops (microservice demonstration) for using it as a teaching materials.

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Our company

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Purpose

Our purpose is to make the learning curve easier for the microservice patterns.

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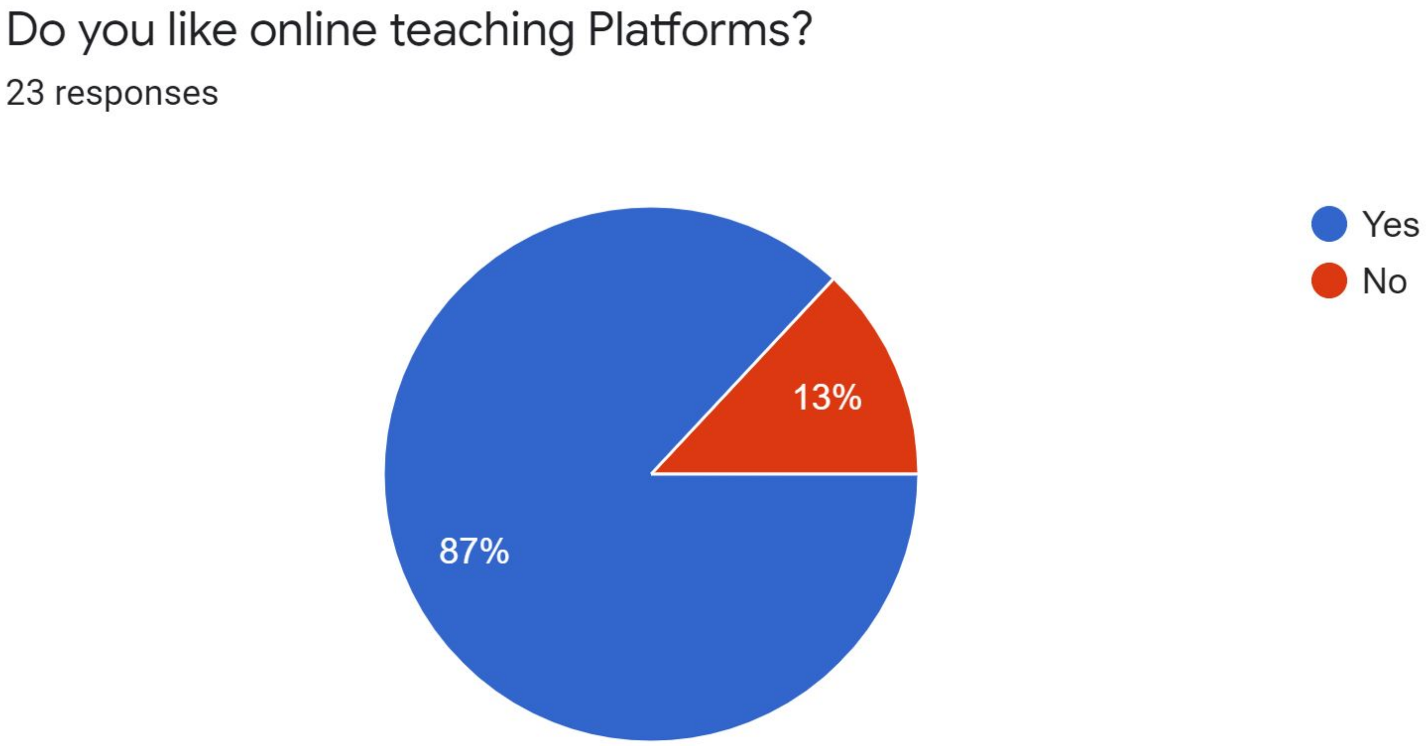
Problem

- Complexity and Long Learning Curve
    - No step by step learning Process
        - Lack of Support

Survey



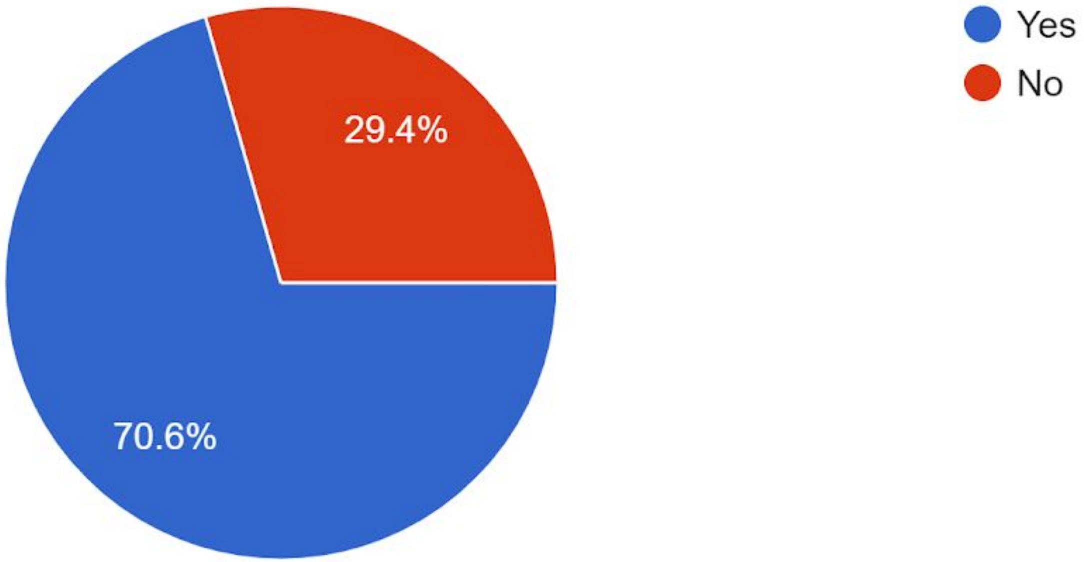
- https://bit.ly/2NQxc02

Survey Result

Topic: Information of the topic ?

Do you know about the Microservices architecture patterns and wanted to learn more?

17 responses

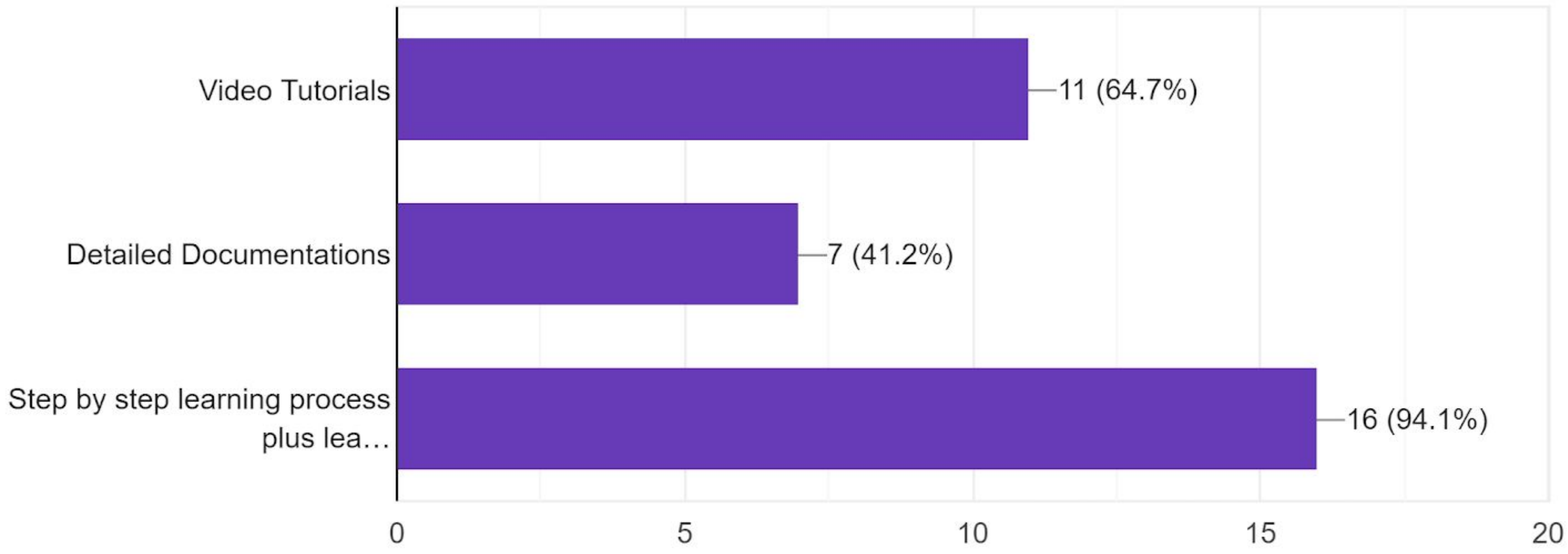


Survey Result

Topic: What they expect from education provider?

What kind of resources do you expect while learning Microservices?

17 responses



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Solution

- Step by step learning process
- Demonstrator
- Interactive Learning tools

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Product

NULLx Learning Kit

Homepage

Learning Microservice With NULLx

Micro service basic course:

Key Features of this Microservices Course:

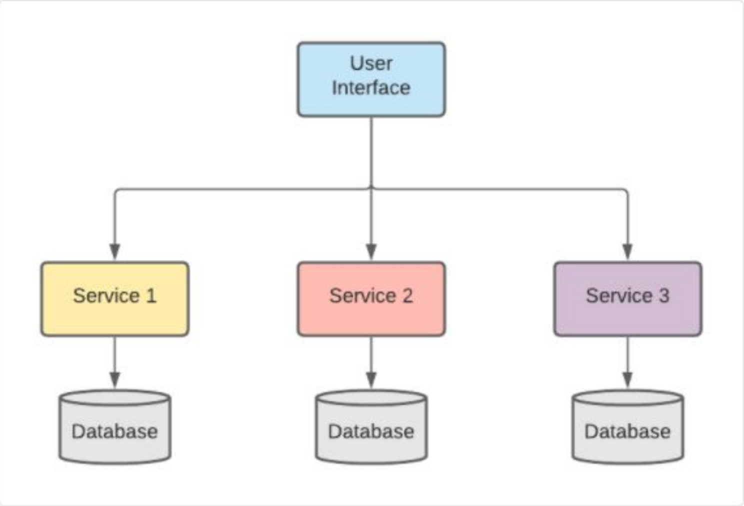
After-course instructor coaching benefit Learning with interactive tools and playable environment

You Will Learn:

Most common pattern of microservice and its architecture. Identify the characteristics of popular microservices, and understand the design differences. Difference between a monolithic application on a single server vs containerized application on multiple cloud instances. Build a simple single-purpose serverless application. Expose an Application Program Interface for the application. Various approaches to infrastructure used in deploying microservices Monitor and maintain microservices in large ecosystems and in the cloud.

What is Microservice:

A microservices architecture an approach and extends it to the loosely coupled services which can be developed, deployed, and maintained independently. Each of these services is responsible for discrete task and can communicate with other services through simple APIs to solve a larger complex business problem.



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Live at

www.nullx-de.github.io/home/NULLxUI

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Demonstration (WebShop)

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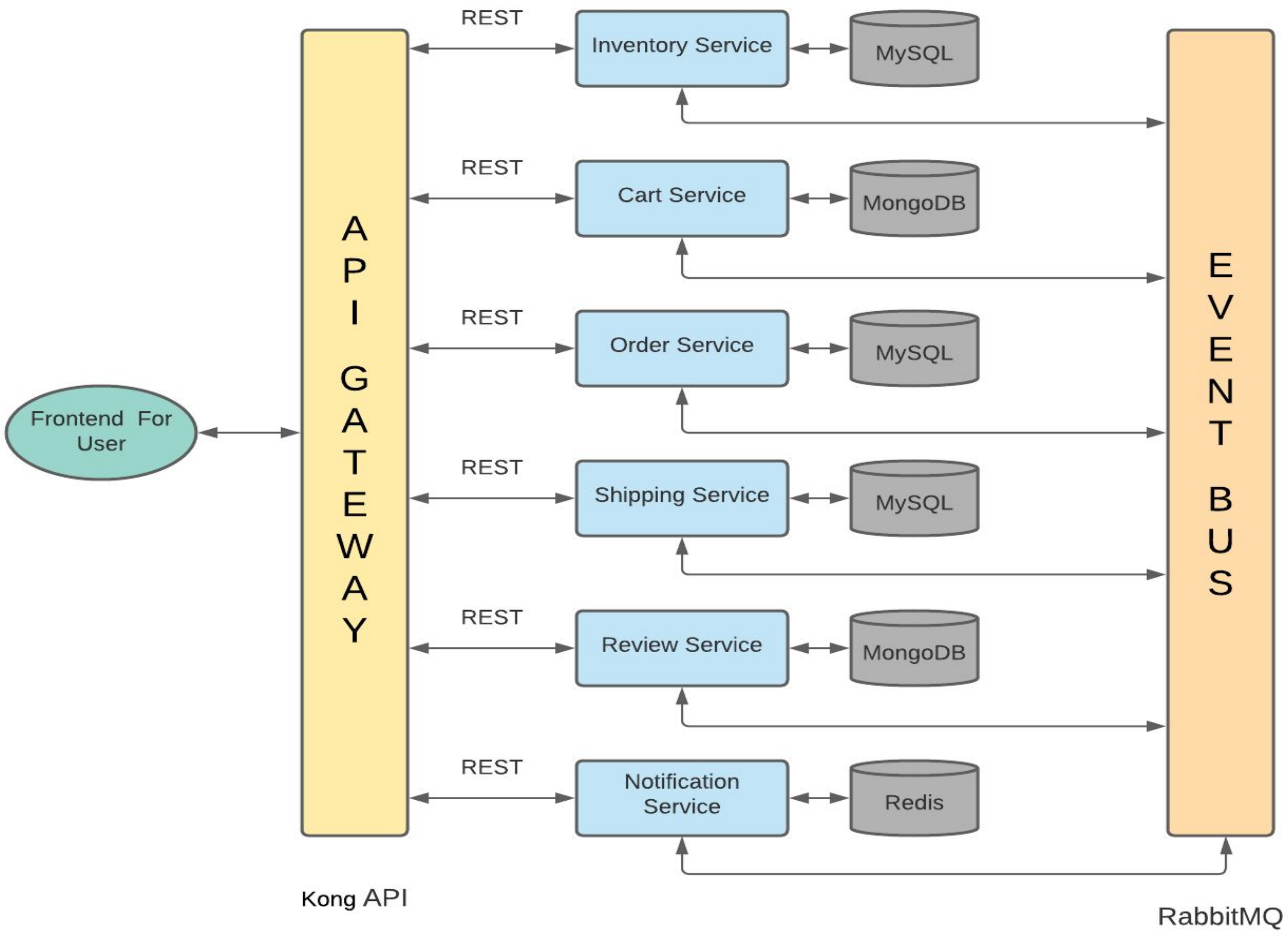
Architecture



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Microservice Patterns

- Domain-Driven Design Pattern Define services corresponding to Domain-Driven Design (DDD) subdomains microservice pattern.
- Database per service Pattern Most services need to persist data in some kind of database.
- Saga Pattern A saga is a sequence of local transactions. Each local transaction updates the database and publishes a message or event to trigger the next local transaction in the saga.
- API Gateway / Backends for Frontends Pattern To give the clients of a Microservices-based application access the individual services.
- Service Instance per container pattern All the services as a (Docker) container image and deploy each service instance as a container in the Kubernetes.

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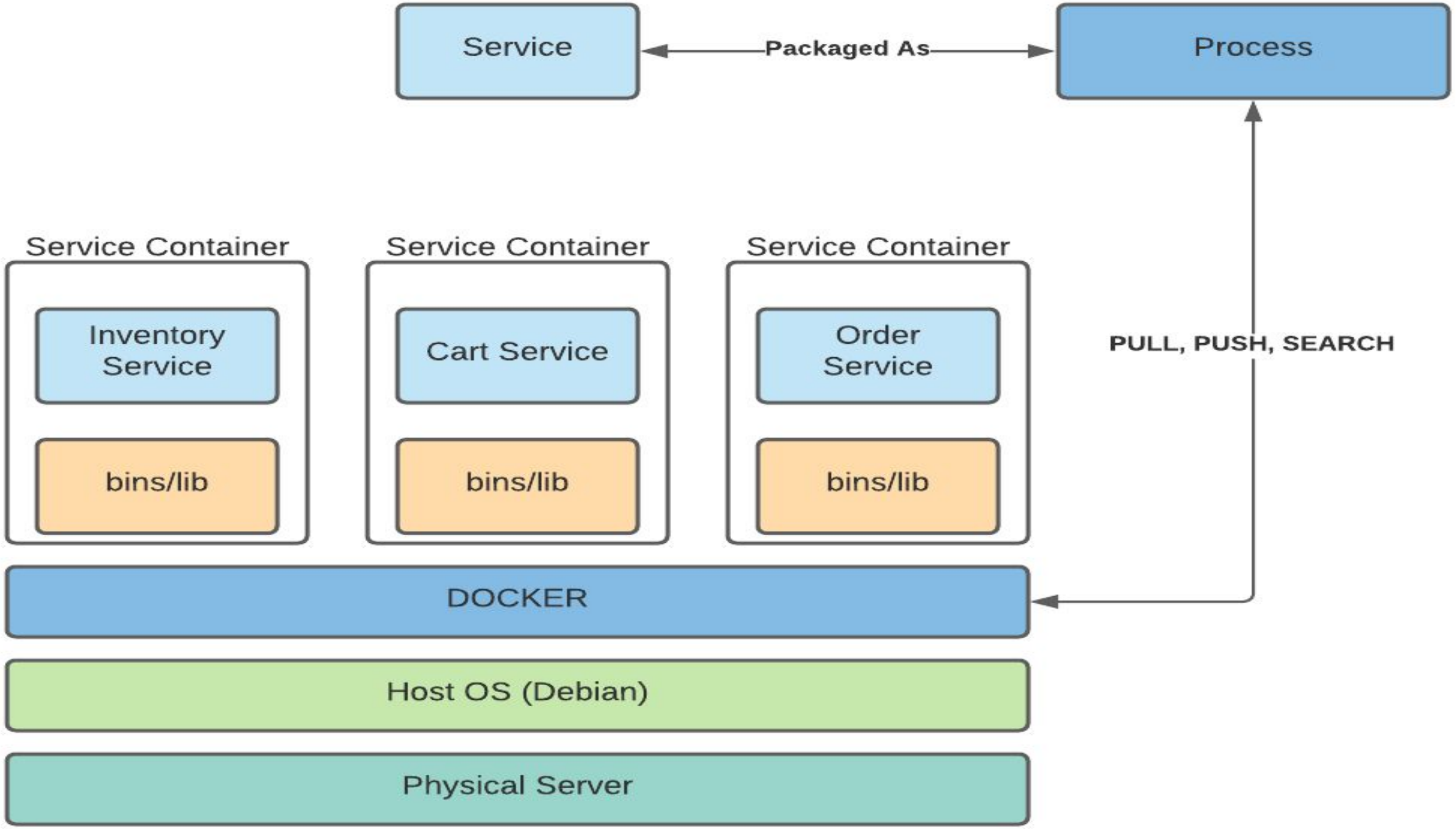
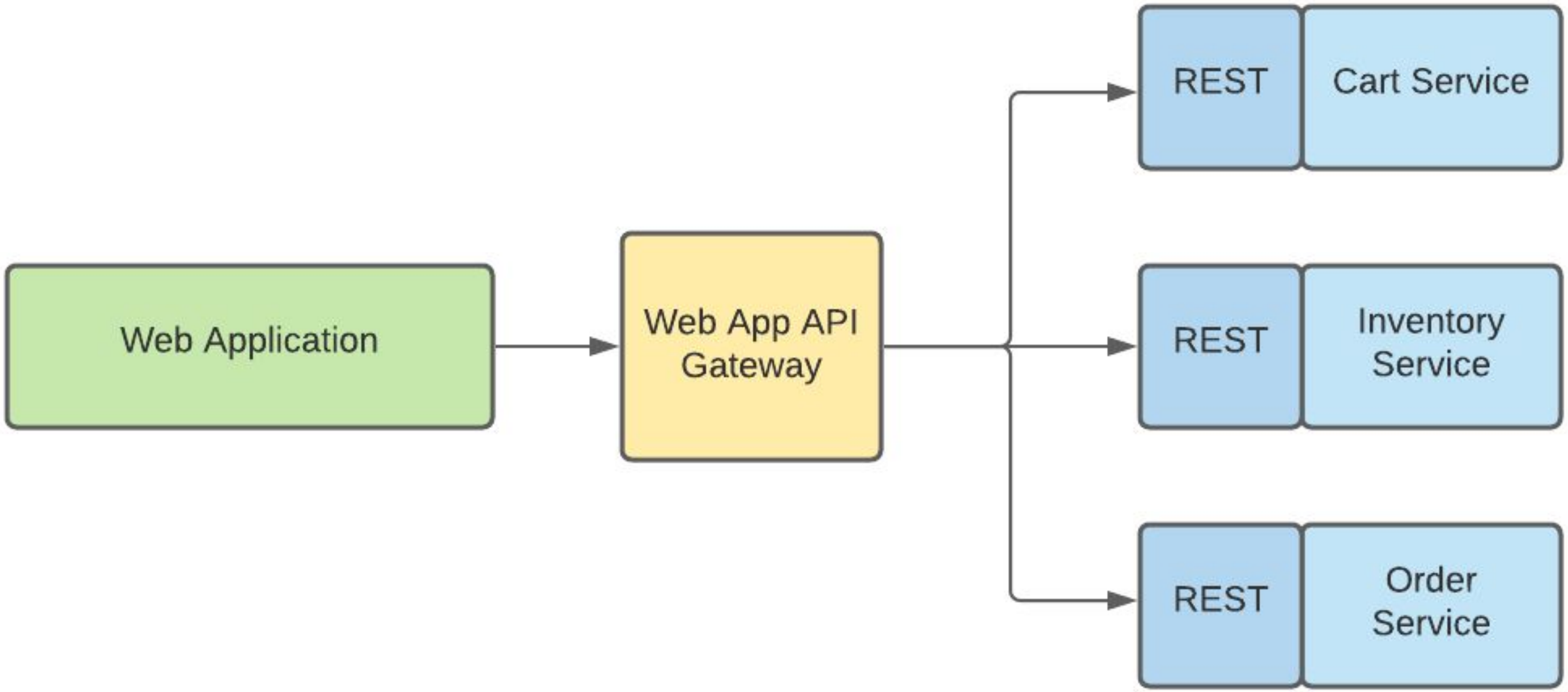
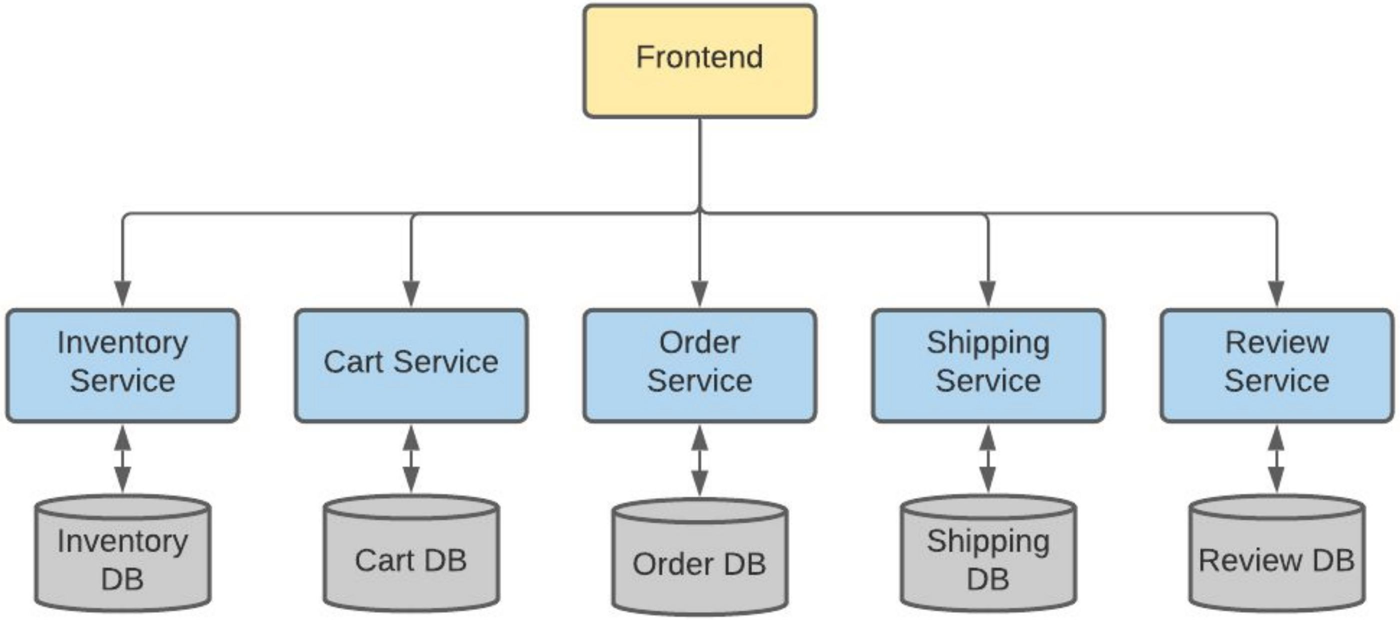
Service Instance Per Container Pattern



Implementing API Gateway Pattern

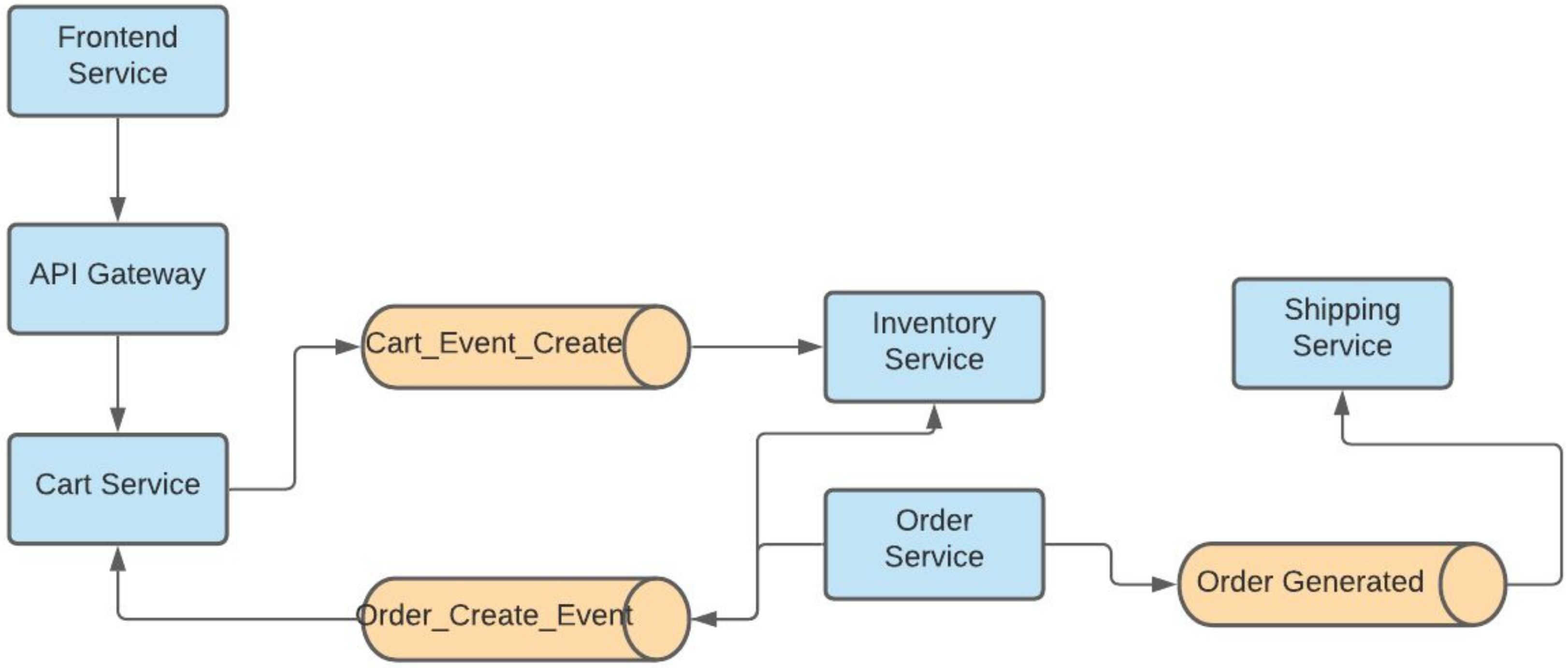


Implementing Database per Service Pattern



- Single Server Instance
- Multiple Database per microservice

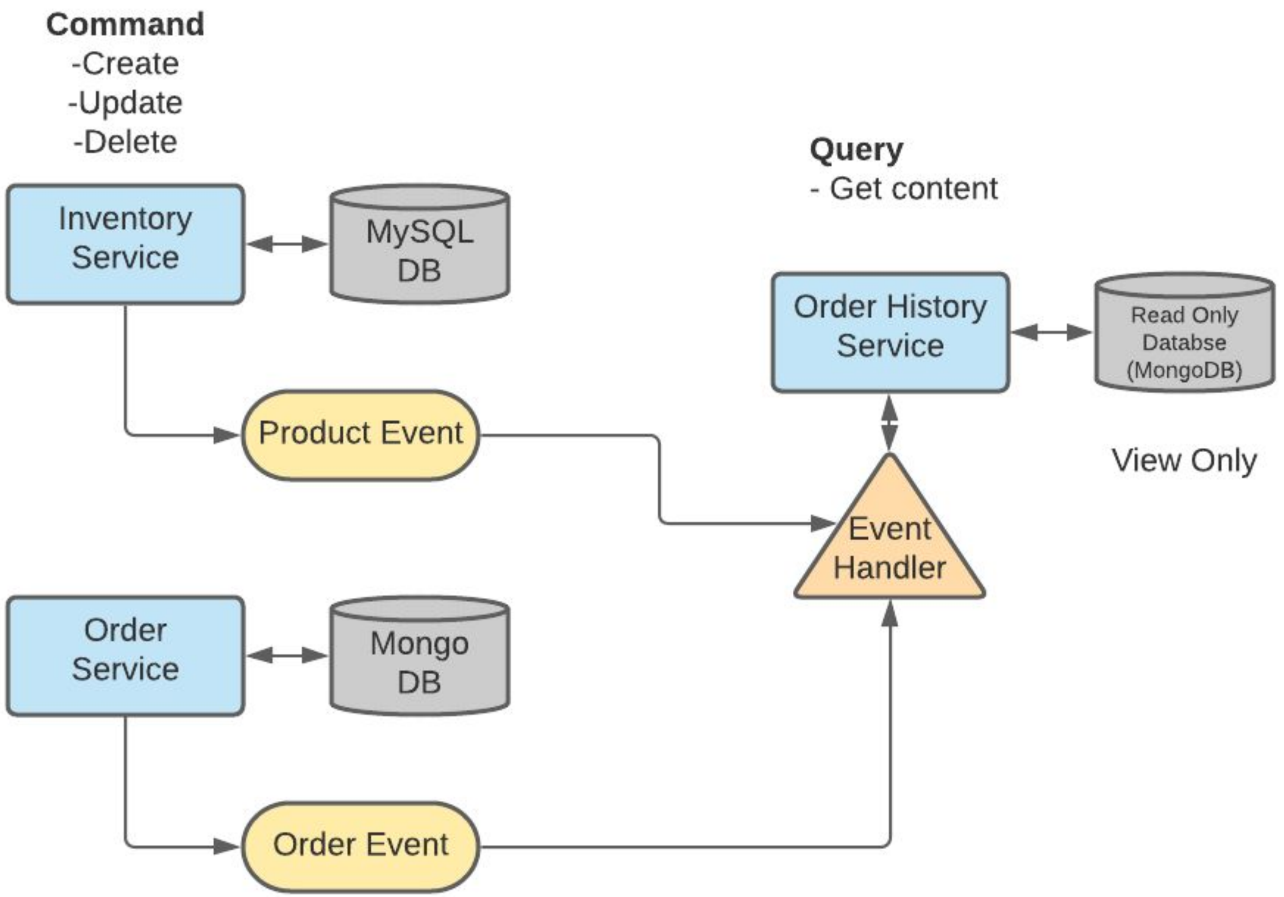
Implementing SAGA Pattern



Updating loosely coupled Database:
Updating Inventory database without ACID transaction by event command.

Error handling:
A product stock out (quantity changed) handled by SAGA choreography.

Implementing Command Query Responsibility Segregation (CQRS) Pattern



Live Demonstration (WebShop)

http://vsr-kub005.informatik.tu-chemnitz.de:30002/

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Gitlab CI/CD

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THANK YOU!

Any Questions ?

DIWAKAR DANGAL

MESHU DEB NATH

RAKIBUL ISLAM

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Appendix

Survey link : https://bit.ly/2NQxc02

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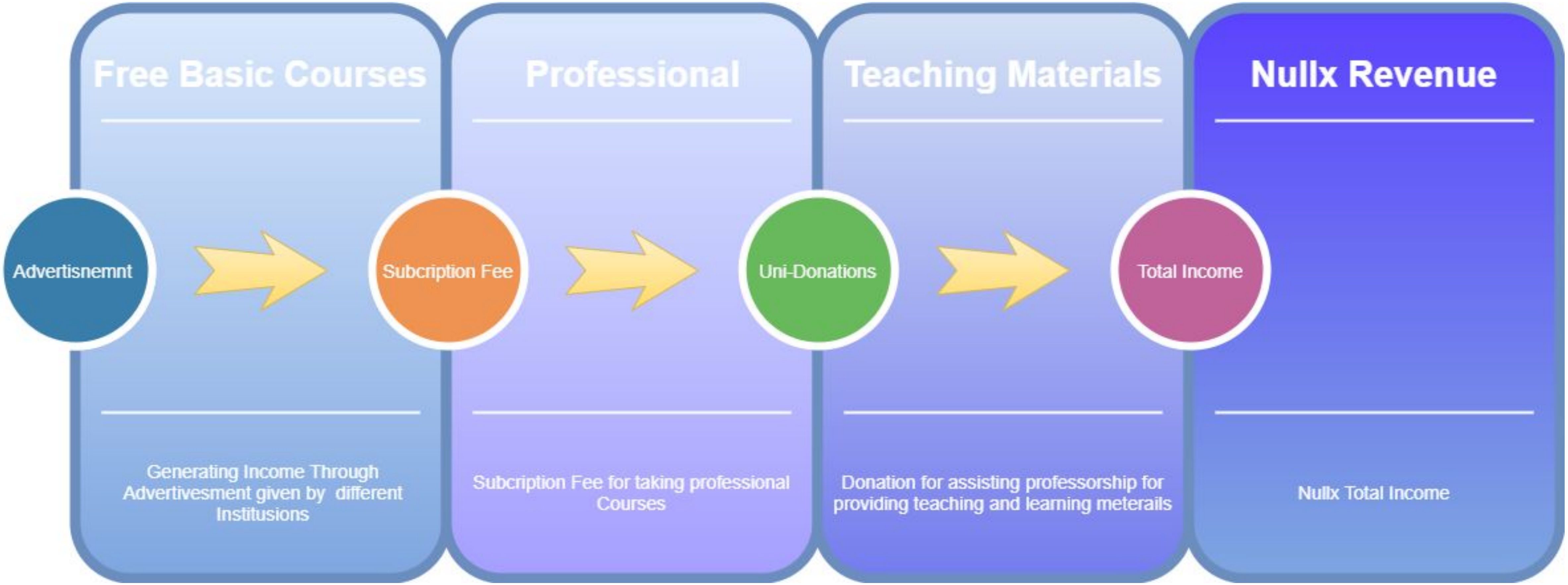
Target Market

Students

Professionals

Educational Institutions

Revenue Stream



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Course subscription

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Teaching and learning material charges